Software Requirements Specification (SRS)

Project: Event Management System

Authors: Vineet Toshniwal | Vankadara Sri Kruthi | Vikhyath Agrawal | Divyanshu Jha

SRN	**Name**
PES2UG24CS589	Vineet Toshniwal
PES2UG24CS574	Vankadara Sri Kruthi
PES2UG24CS585	Vikhyath Agrawal
PES2UG24CS915	Divyanshu Jha

Revision History

Version	Date	Author	Change Summary	Approval Status
1.0	05-09-2026	Vineet Toshniwal	Initial SRS prepared for Event Management System	Pending
1.0	05-09-2026	Vankadara Sri Kruthi	Initial SRS prepared for Event Management System	Pending
1.0	05-09-2026	Vikhyath Agrawal	Initial SRS prepared for Event Management System	Pending
1.0	05-09-2026	Divyanshu Jha	Initial SRS prepared for Event Management System	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	<Name>	<Signature / Email>	<Date>
Project Guide	<Name>	<Signature / Email>	<Date>

Table of Contents

1. Introduction

2. Overall Description

3. External Interface Requirements

4. System Features (Detailed)

5. Non-Functional Requirements (Detailed)

6. Quality Attributes & Acceptance Tests

7. UML Use-Case Diagrams

8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Event Management System. It defines the functional and non-functional requirements, interfaces, security requirements, quality attributes, and verification criteria for a system that supports planning, organizing, registering, conducting, and monitoring events.

1.2 Scope

The Event Management System covers event creation and scheduling, venue management, organizer and attendee management, event registration, ticketing, payment recording, notifications, attendance tracking, feedback, reporting, and administrative monitoring. It excludes the internal implementation of external payment gateways, email/SMS providers, and third-party authentication services.

1.3 Audience

The intended audience includes developers, QA engineers, project guides, administrators, event organizers, event staff, attendees, and assessment evaluators.

1.4 Definitions

EMS – Event Management System; Admin – system administrator; Organizer – person responsible for creating and managing an event; Attendee – registered participant; Venue – physical or virtual location where an event is conducted; RSVP – response indicating attendance; UI – User Interface; API – Application Programming Interface; OTP – One-Time Password; QR – Quick Response code.

2. Overall Description

2.1 Product Perspective

The Event Management System is a web-based application that provides a centralized platform for organizers and administrators to manage events and for attendees to discover, register for, and participate in events. The system interacts with a database and may integrate with external payment and notification services.

2.2 Major Product Functions

- User registration, login, and role-based authentication
- Create, edit, publish, cancel, and manage events
- Manage venues, schedules, capacity, and event categories
- Search, filter, and view available events
- Register attendees for events and manage participant lists
- Generate and validate event tickets or QR-based passes
- Record and verify online/offline payment status
- Send registration, reminder, update, and cancellation notifications
- Track attendance and check-in/check-out
- Collect attendee feedback and ratings

- Generate event, registration, attendance, and revenue reports
- Provide administrator controls, audit logs, and system monitoring

2.3 User Roles and Characteristics

Role	Characteristics / Responsibilities
Admin	Manages users, events, venues, categories, system settings, reports, and audit information.
Organizer	Creates and manages events, schedules, registrations, tickets, attendance, and event reports.
Attendee	Browses events, registers, makes payments where required, receives tickets, checks in, and submits feedback.
Event Staff	Verifies tickets/QR codes, records attendance, and assists with event operations.

2.4 Operating Environment

The system shall operate as a responsive web application on modern browsers and shall use a server-side application, relational database, and secure network connection. It should support desktop and mobile browsers.

2.5 Constraints

- The system shall protect personal and payment-related information.
- External payment and notification services may be unavailable and shall be handled gracefully.
- Only authorized users shall access administrative and organizer functions.
- Event registration shall respect configured venue capacity and registration limits.
- The system shall maintain consistent event, registration, payment, and attendance records.

3. External Interface Requirements

3.1 User Interfaces

The primary interface shall be a responsive web UI with dashboards appropriate to each user role. Forms shall provide validation messages, confirmation messages, and clear error handling. Event pages shall display title, description, date, time, venue, capacity, registration status, and applicable fee.

3.2 Hardware Interfaces

No specialized hardware is mandatory. The system may use a smartphone or computer camera for QR-code scanning and may optionally support printers for tickets or reports.

3.3 Software Interfaces

- Relational Database Management System for persistent storage.
- Payment Gateway API for processing online payments where enabled.
- Email/SMS/notification API for event notifications and reminders.

- Optional authentication provider API for secure login.

3.4 Communications

All client-server communication shall use HTTPS/TLS. API requests shall use authenticated sessions or tokens where applicable. The system shall handle network failures without corrupting event or registration data.

Template note: the reference SRS requires at least 15 functional requirements, 5 non-functional requirements, 2 security objectives, 5 security requirements, and at least 2 UML use-case diagrams.

4. System Features (Detailed)

Each requirement below includes a requirement ID, type, priority, stakeholder/source, acceptance criteria, and test-case reference.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria	Test Ref
EMS-F-001	The system shall allow users to create an account using required personal details and credentials.	Functional	High	Attendee / Organizer	Valid data creates an account and confirmation is displayed.	TC-REG-01
EMS-F-002	The system shall authenticate users and provide access according to their assigned role.	Functional	High	All users	Valid credentials lead to the correct role dashboard; invalid credentials are rejected.	TC-AUTH-01
EMS-F-003	The system shall allow organizers to create an event with title, description, date, time, venue, capacity, category, and registration details.	Functional	High	Organizer	A valid event is saved with all mandatory fields.	TC-EVT-01

EMS-F-004	The system shall allow organizers to edit, publish, postpone, or cancel events subject to authorization.	Functional	High	Organizer	Changes are saved and affected users can see the updated event status.	TC-EVT-02
EMS-F-005	The system shall allow attendees to search and filter events by category, date, venue, and availability.	Functional	Medium	Attendee	Search and filters return matching events.	TC-SEARCH-01
EMS-F-006	The system shall display complete event details including schedule, venue, capacity, fee, and registration status.	Functional	High	Attendee	Event page shows current and accurate information.	TC-EVT-03
EMS-F-007	The system shall allow an eligible attendee to register for an event if seats are available.	Functional	High	Attendee	Successful registration creates a unique registration record and reduces available capacity.	TC-REG-02
EMS-F-008	The system shall prevent registrations when an event reaches its configured capacity.	Functional	High	Organizer / Attendee	Registration is rejected when no seats remain and the event is marked full.	TC-REG-03
EMS-F-009	The system shall record payment status for events that require a registration fee.	Functional	High	Attendee / Organizer	Successful payment changes the registration status to paid; failed payment does not confirm registration.	TC-PAY-01

EMS-F-010	The system shall generate a unique digital ticket or QR-based pass for a confirmed registration.	Functional	High	Attendee	Confirmed registration produces a unique ticket/pass.	TC-TKT-01
EMS-F-011	The system shall allow event staff to validate a ticket or QR code during check-in.	Functional	High	Event Staff	A valid unused pass is accepted and marked checked-in; invalid or reused passes are rejected.	TC-CHK-01
EMS-F-012	The system shall send registration, reminder, schedule-change, and cancellation notifications to affected users.	Functional	Medium	Attendee / Organizer	A configured notification is generated after the triggering event.	TC-NOTIF-01
EMS-F-013	The system shall allow organizers and administrators to view attendee lists and registration status.	Functional	High	Organizer / Admin	Authorized users can view and filter participant records.	TC-LIST-01
EMS-F-014	The system shall allow attendees to submit feedback and ratings for eligible completed events.	Functional	Medium	Attendee	A registered attendee can submit one valid feedback response after the event.	TC-FB-01
EMS-F-015	The system shall generate reports on registrations, attendance, payments, and event	Functional	High	Organizer / Admin	Authorized users can generate reports for selected events and date ranges.	TC-REP-01

	performance.					
EMS-F-016	The system shall allow administrators to manage users, event categories, venues, and system configuration.	Functional	High	Admin	Authorized administrator changes are saved and reflected in the system.	TC-ADMIN-01
EMS-F-017	The system shall maintain an audit record of important administrative and event-management actions.	Functional	High	Admin	Create/update/delete/status-change actions record actor, timestamp, and action details.	TC-AUD-01

5. Non-Functional Requirements (Detailed)

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement	Test Ref
EMS-NF-001	Normal user actions shall receive a server response within 3 seconds for at least 95% of requests under expected load.	Performance	High	95th percentile response time ≤ 3s in load testing.	TC-PERF-01
EMS-NF-002	The system shall provide at least 99.5% monthly availability excluding scheduled maintenance.	Reliability	High	Monthly uptime report shows ≥99.5%.	TC-REL-01
EMS-NF-003	The system shall protect personal data and shall not store payment card credentials in	Security / Compliance	High	Security review confirms sensitive data is protected and card	TC-SEC-01

	the application database.			credentials are not stored.	
EMS-NF-004	The system shall support at least 500 concurrent users under the defined deployment configuration without critical failure.	Scalability	Medium	Load test completes successfully for 500 concurrent users.	TC-SCALE-01
EMS-NF-005	The interface shall be usable on current desktop and mobile browsers and provide clear validation and error messages.	Usability	Medium	Usability test passes on supported browsers and screen sizes.	TC-UX-01
EMS-NF-006	The system shall maintain timestamped logs for security, administrative, registration, payment-status, and attendance events.	Auditability	High	Required events are present with timestamp and actor/system source.	TC-OPS-01

5.1 Security

5.1.1 Security Objectives

- Protect confidentiality and integrity of user, event, registration, and payment-related information.
- Ensure that only authenticated and authorized users can perform privileged operations.
- Maintain accountability through secure logging of important system and administrative actions.

5.1.2 Security Requirements

Req ID	**Requirement**	**Type**	**Priority**	**Acceptance Criteria / Test Case**
EMS-SR-001	All client-server communications shall use HTTPS with TLS 1.2 or higher.	Security	High	Network inspection confirms encrypted transport.

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test Case	Test Ref
EMS-SR-001	All client-server communications shall use HTTPS with TLS 1.2 or higher.	Security	High	Network inspection confirms encrypted transport.	TC-SEC-01
EMS-SR-002	Passwords shall be stored only as strong salted password hashes and shall never be stored in plaintext.	Security	High	Database inspection and security test confirm password hashing.	TC-SEC-02
EMS-SR-003	The system shall enforce role-based access control for Admin, Organizer, Attendee, and Event Staff functions.	Security	High	Unauthorized role access is denied.	TC-SEC-03
EMS-SR-004	The system shall validate and sanitize user input to reduce injection and malicious input risks.	Security	High	Security testing shows malicious test inputs are rejected or safely handled.	TC-SEC-04
EMS-SR-005	The system shall expire inactive authenticated sessions after a configured timeout and shall invalidate logout sessions.	Security	Medium	Expired sessions cannot access protected resources.	TC-SEC-05
EMS-SR-006	Important security and administrative events shall be logged without storing passwords or other unnecessary secrets.	Security	High	Audit-log review confirms required events and absence of credentials.	TC-SEC-06

6. Quality Attributes & Acceptance Tests

- Functional completeness: all high-priority functional requirements are implemented and verified.
- Performance: response-time and concurrency targets are met under production-like test conditions.
- Security: all mandatory security requirements pass security testing.
- Reliability: the system meets the stated availability and error-handling requirements.
- Usability: representative users can complete registration, ticket retrieval, and event-management tasks without critical usability issues.

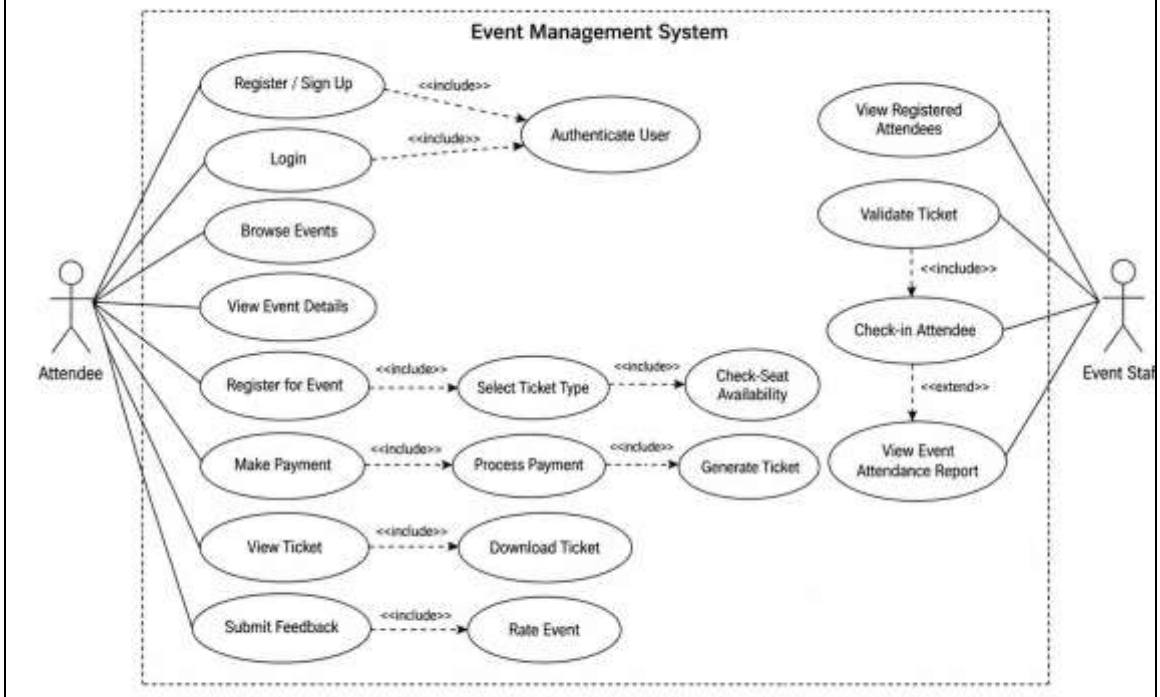
Acceptance exit criteria: all high-priority functional requirements are implemented and verified, no critical security or non-functional requirement failures remain, and the RTM contains test coverage for all requirements.

Acceptance Test Suites

Suite	Coverage
Authentication	Registration, login, logout, role-based access
Event Management	Create, edit, publish, cancel, and schedule events
Registration & Capacity	Registration, duplicate handling, capacity limits
Payment & Ticketing	Payment status, confirmation, ticket/QR generation
Attendance	QR validation and check-in
Notifications	Registration, reminder, update, cancellation notifications
Reports	Registration, attendance, payment, and performance reports
Performance & Security	Response time, concurrency, access control, input validation, secure transport

7. System Models and Diagrams

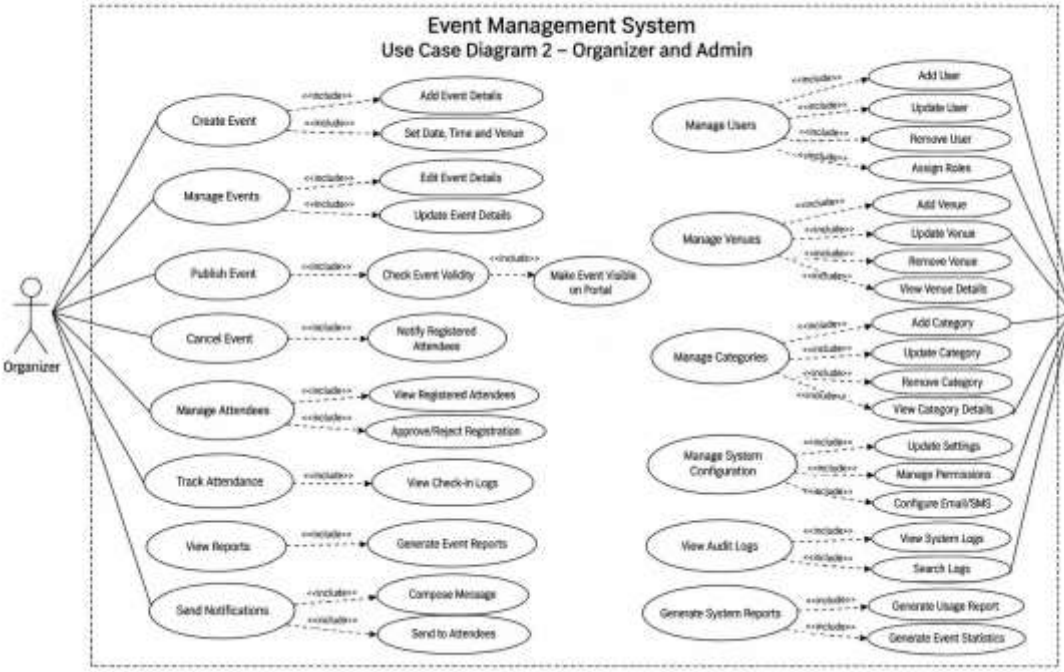
7.1 UML Use-Case Diagram 1 – Attendee and Event Staff



Actors: Attendee, Event Staff. Main use cases: Register/Login, Browse Events, View Event Details, Register for Event, Make Payment, Receive Ticket/QR Pass, View Registration, Submit Feedback, Validate Ticket, Check In Attendee.

Suggested relationships: Register for Event includes Check Availability; Paid Registration includes Make Payment; Check In Attendee includes Validate Ticket.

7.2 UML Use-Case Diagram 2 – Organizer and Admin



Actors: Organizer, Admin. Main use cases: Create Event, Edit Event, Publish Event, Cancel Event, Manage Venue, Manage Attendees, Track Attendance, View Reports, Send Notifications, Manage Users, Manage Categories, Manage System Configuration, View Audit Logs.

At least two UML use-case diagrams should be created, following the reference template requirement. These textual definitions can be used directly to draw the diagrams.

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)
EMS-F-001	User registration	4 / DS-Auth-01	Auth Module	TC-REG-01	N
EMS-F-002	User authentication	4 / DS-Auth-02	Auth Module	TC-AUTH-01	N

EMS-F-003	Create event	4 / DS-Event-01	Event Module	TC-EVT-01	N
EMS-F-007	Event registration	4 / DS-Reg-01	Registration Module	TC-REG-02	N
EMS-F-009	Payment status	4 / DS-Pay-01	Payment Module	TC-PAY-01	N
EMS-F-010	Ticket/QR generation	4 / DS-Ticket-01	Ticket Module	TC-TKT-01	N
EMS-F-011	Attendance check-in	4 / DS-Attendance-01	Attendance Module	TC-CHK-01	N
EMS-F-015	Reports	4 / DS-Report-01	Reporting Module	TC-REP-01	N
EMS-NF-001	Response time	5 / DS-Perf-01	Web/API Layer	TC-PERF-01	N
EMS-NF-003	Data protection	5 / DS-Sec-01	Security Layer	TC-SEC-01	N
EMS-SR-001	TLS 1.2+	5.1 / DS-Sec-02	Network/API Layer	TC-SEC-01	N
EMS-SR-003	Role-based access	5.1 / DS-Sec-03	Authorization Module	TC-SEC-03	N

End of Software Requirements Specification